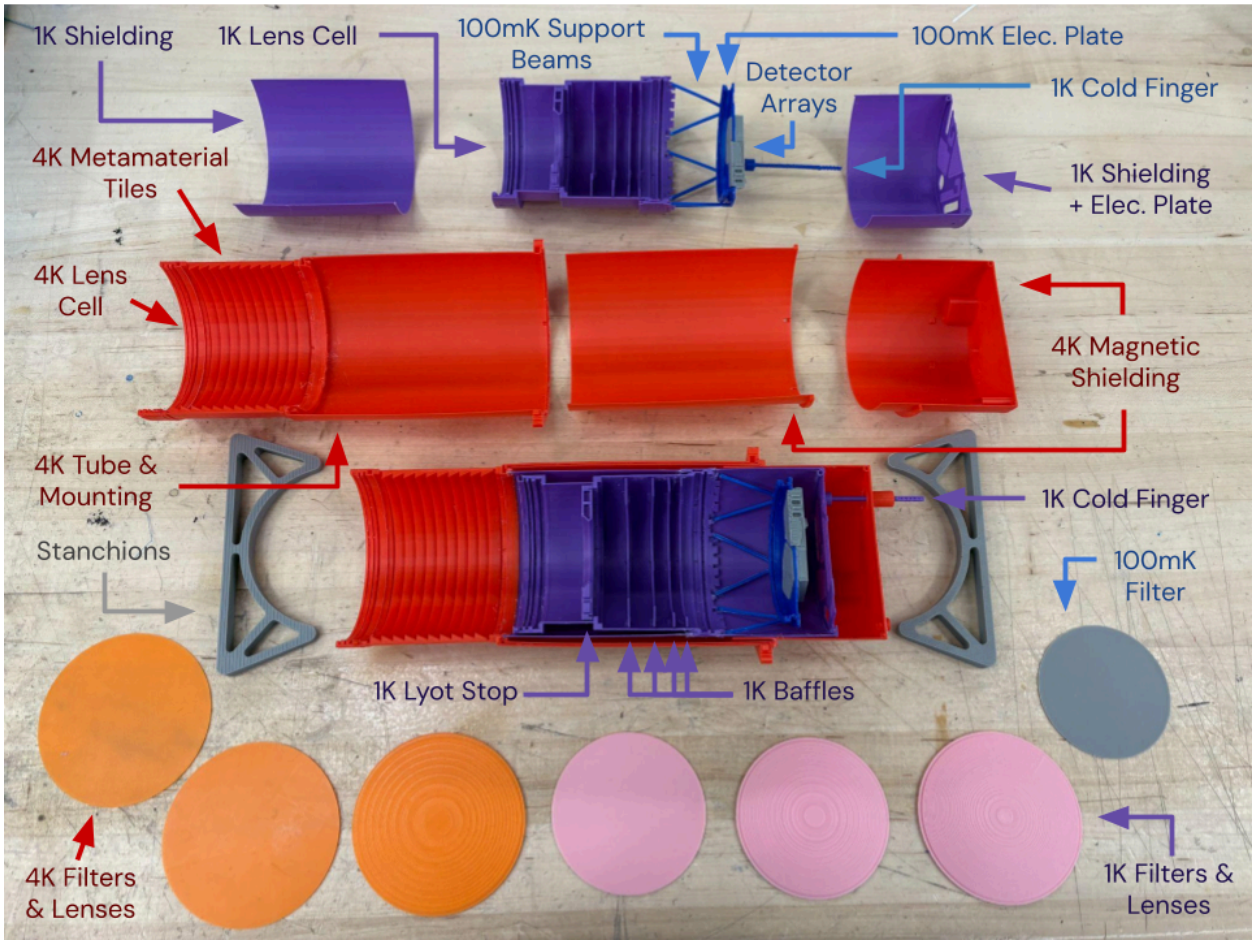
Optics Tube ¼-Scale Model Assembly Guide

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Printing

3D Printing Components can be done with any printer with an extrusion diameter of 0.4mm or less.

Colors were chosen to demonstrate temperature stages in the instrument:

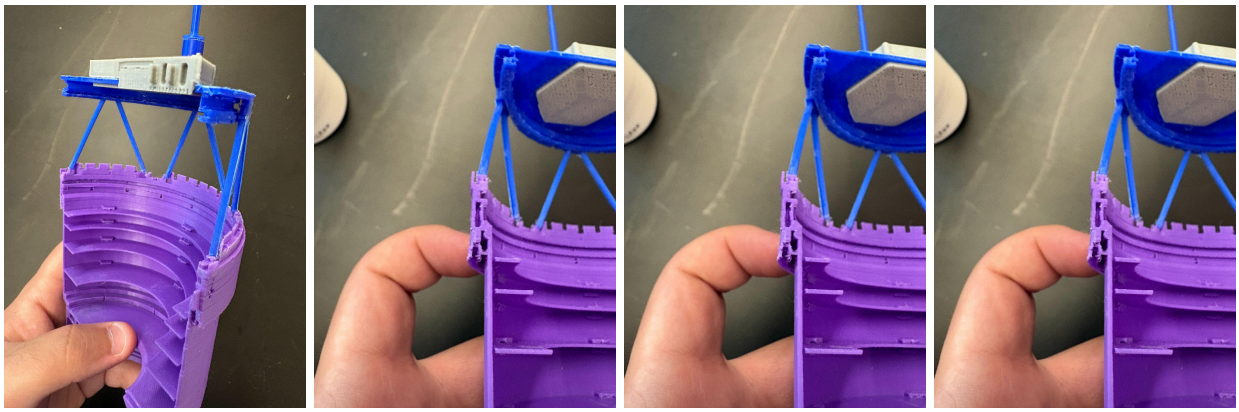
Stage	**Primary Color** (tubing, shielding, lens cells, other misc. components)	**Secondary Color** (Lenses, filters, detectors)
40K	**N/A** (Prime-Cam only; Module itself held at 4K)	
4K	Red	Orange
1K	Purple	Pink
100mK	Blue	Gray

STLs of the model, divided by temperature, are included in the GitHub.

Some components may need more support than others. Some may benefit greatly from the presence of a brim, while others should not be printed with one (Lenses and Filters, misc. components of the end 4K Shielding). The cold fingers should be printed with other items so as to avoid distortion due to flexure during printing.

Pre-Assembly Info

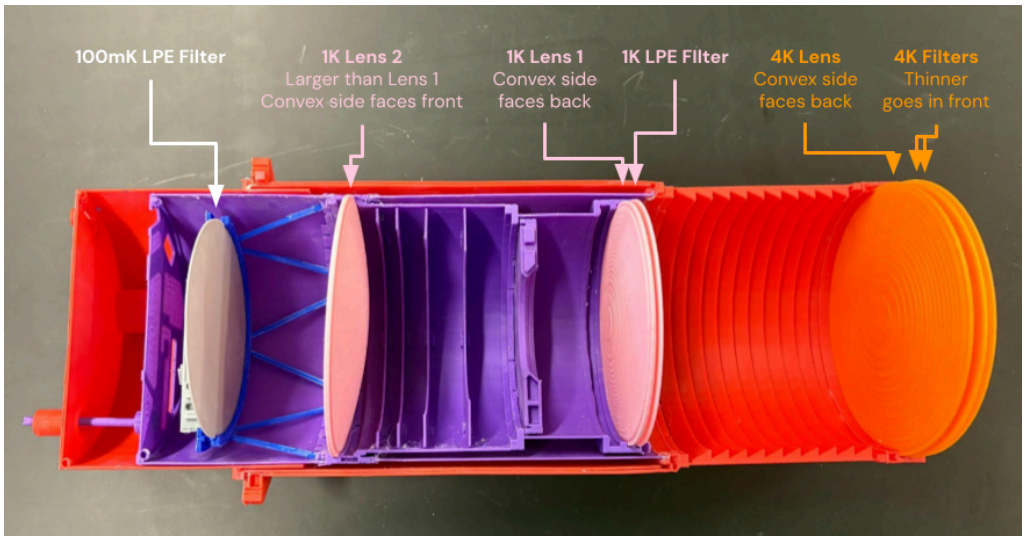
The model holds itself together via slots on various parts. Thin parts, representing the magnetic shielding, are inserted into slots to support inner components (this is how the real instrument is assembled). Many of these slots and shielding components have interlocking tabs, which prevent parts from rotating or sliding out when the model is assembled and/or stored horizontally. The tabs are bilaterally symmetric, meaning that components are *not* identical. It is therefore important to make sure the tabs are all aligned and properly fitted when assembling, especially if/when parts are glued together.



There are also larger tabs on top of the **1K Lens Cell** to hold the **1K Magnetic Shielding** as a slot would.

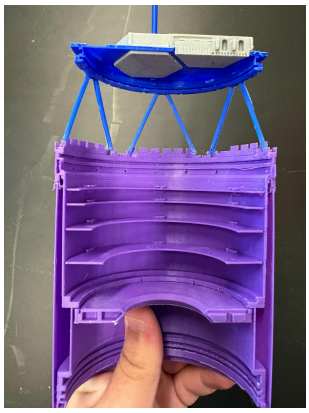
For the purposes of this document, the terms **front and bottom** refer to the part of the model closest to the **4K Lens Cell**, and the terms **back and top** refer to the part of the model farthest.

Arrangement of lenses



Gluing

IMPORTANT NOTE: Components should not be glued across the center dividing line.



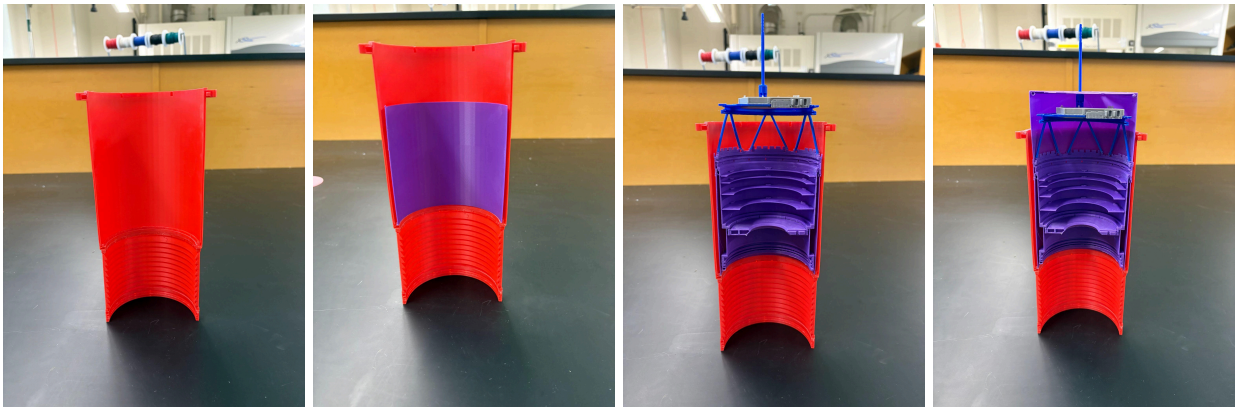
The following components should be glued together

- Each half of the **4K Lens Cell, Metamaterial Tiles, and Tube & Mounting Structure** Should be glued
- The miscellaneous components of the end **4K Magnetic Shielding** (**Cold Finger Tubes, Rear Cover,** and other attachments) printed separately.
- The **Baffles** and **Lyot Stop** should be glued to each half the 1K Lens Cell.
    - Starting with the one closest to the **Lyot Stop**, the **Baffles** should be ordered from thickest to thinnest.
- Each **Cold Finger** should be glued to its respective **Electrical Plate**.
    - The easiest way to do this is to glue the provided **Connector** in place first, wait for it to dry, and then use it to align the **Cold Finger** when gluing that component.
- The six **100mK Support Beam** structures should be glued to the six spots on the **100mK Electrical Plate** (three on each side).
- The two halves of the **Detector Arrays** should be glued to each half of the **100mK Electrical Plate**. Make sure the **Detector Arrays** are facing the right direction.
- The shorter **1K Magnetic Shielding** halves should be glued to the **1K Electrical Plate** halves, using the ring structure on the latter to align the former.
    - Pay attention to where the tabs on the **Shielding** are going.
        - You'll want to keep them aligned properly so the model can close and stay in place.
        - Use what components are already in place as a guide for which half to use to keep the tabs aligned.
- Once everything else is dry, glue the completed **100mK Section** to the top of the **1K Lens Cell**.
    - Each **Beam** should be glued to the angled bracket on the **1K Lens Cell**.
    - Be prepared to hold these in place so they can dry.

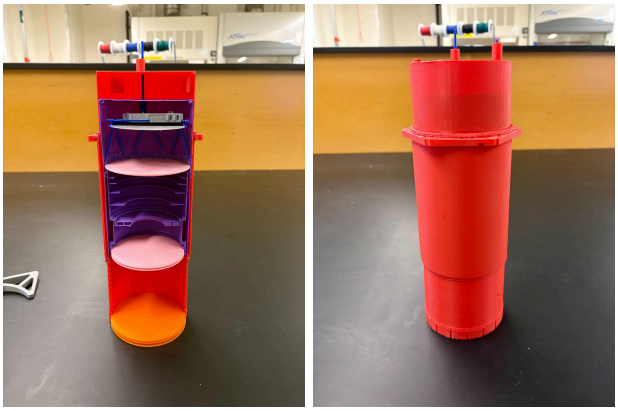
To aid stability and/or assembly, additional gluing can be done between components, including gluing one or both halves together, lenses to cells, or temperature stages together.

Assembly

The following steps should be taken to assemble the model



- Lay one half of the **4K Tube** (the three glued 4K components) *vertically*, with the **Lens Cell** on the bottom.
- Place the unglued **1K Shielding** into the slot on the lower **4K Tube** (the three glued 4K components). Place the **1K Lens Cell** on top of that. Make sure the tabs are alinged
- Place the **1K Shielding + Electrical Plate** on top of the **1K Lens Cell**, making sure the **Shielding** is inside the three large, vertical tabs.
- Slide the middle **4K Shielding** between the **4K Tube** and **1K Lens Cell**. Make sure that the holes in the **4K Shielding** and **Mounting Structure** are aligned.
- Place the end **4K Shielding** on top of the exposed end of the middle **4K Shielding**, such that the former's tabs go into the latter's holes.
- Repeat this process with the other half
- Insert all **lenses** and **filters** into their proper locations and orientations
- Bring the two halves together.

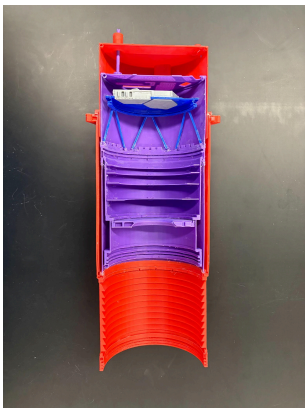
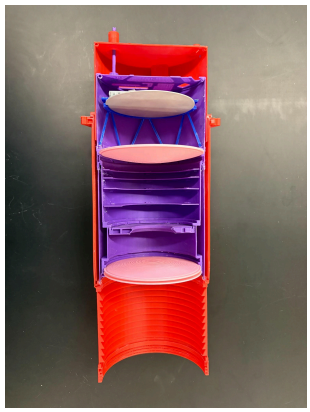
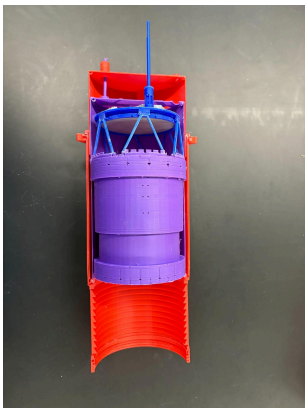
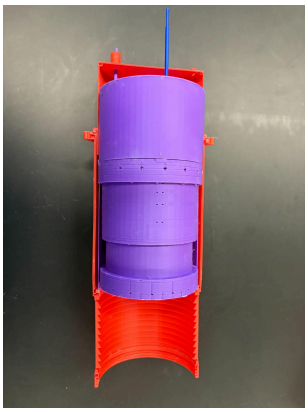


Alternate Assembly
If one of the halves is fully glued together, this alternate method can be used to make assembly easier and quicker.

The method can be made easier by making the following modifications to the **unglued half components**:

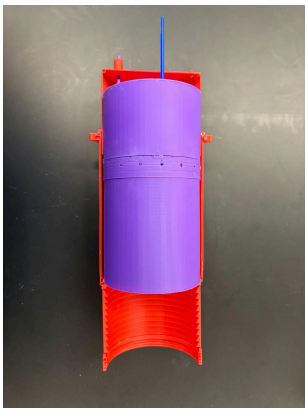
- Remove the central large tab on the **1K Lens Cell.**
- Remove the *inner* edge of the slot on the **4K Tube & Mounting Structure.**
  - Remove the *outer* edge of the bottom slot on the **1K Lens Cell.**
- Cut the tabs off of the end **4K Shielding.**

The assembly steps:

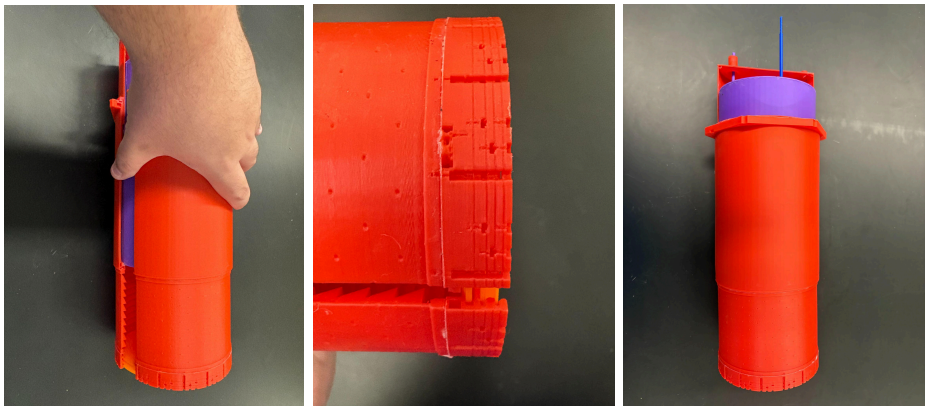





- Place the **half-model** *horizontally*.
- Insert the **1K and 100mK lenses and filters** into their proper locations on the **half-model**.
- Attach the **1K Lens Cell** to the **half-model**. Make sure the large tabs on the **Detector Arrays** are inserted together.
- Lower the **1K Shielding + Electrical Plate** into the upper pins and two remaining tabs.

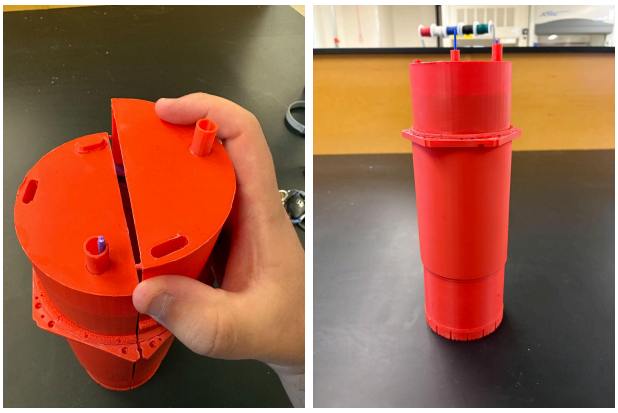





- Slot the lower **1K Shielding** into place.
- Insert the **4K lenses and filters** into their proper locations on the **half-model**.



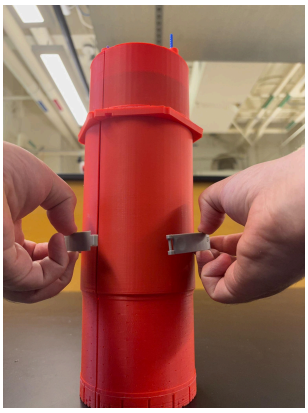
- Attach the lower **4K Tube** to the **half-model**.
    - Make sure the lenses are all in their slots on the top half.
- Orient the model *vertically.*
- Slide the middle **4K Shielding** between the **4K Tube** and **1K Lens Cell**.
- Fit the end **4K Shielding** into place, including the pins.



In both cases, disassemble the model by taking the above steps in reverse.

Additional Components

If needed, one can place the **Clamp Ring** around the model to hold it shut.




- To attach, stretch the ring around the middle tube and click the two ends together.
- To remove, align the clip with the seam of the middle tube. Squeeze the left end until the two halves detach. Alternatively, one can slide the ring down the middle tube to remove it.






The stanchions are also included to help display the model.

- The two stanchions should read **A4K Metamaterial Tiles** and **4K Magnetic Shielding**, respectively.
  - These should tell you where each should be placed for the best fit, although the difference between each is very small.



